Text loops: wells, the birthday law and model menus

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Summary

A story that is summarised to 50 words and rewritten, 100 times over, does not converge the way Hintze's image loops do. It falls into one of a few hundred narrow wells, and which well depends on the model more than on the seed. Twenty cells (five models, 20 or 100 seeds each, about 72,000 stories) give five results.

1. **Wells exist, and they obey the birthday law.** No two of 20 trajectories ever meet. At 100 trajectories, 13 wells are shared, and in a second 100-seed sample 17. The chance that a given pair shares a well is constant (0.006 and 0.005), so sharing grows with the square of the number of trajectories. The wells recur across samples as plot families (7 of 13 revisited) and are stable under projection onto a fixed atlas.
2. **The wells are plots, and the plots belong to the model.** Haiku's wells are domestic object-plots: a shoebox of unopened letters from an estranged relative, a dead relative's ledger of forgiven debts, an unfinished melody. Mistral's and Qwen's wells are civic macro-plots: a stranger revives a decaying town, a neglected lot becomes a community garden. No 7B trajectory enters a Haiku well in 200 tries, while the two 7B models enter each other's.
3. **A one-sentence theme bottleneck removes the plot and keeps the moral.** Under a 25-word theme, the seed is gone in one step and every trajectory re-rolls its plot each step, yet the moral persists: each model settles on a menu of about five morals. Sonnet's menu is Haiku's (0.82 of its stories fit Haiku's five morals), with different weights; the 7B menus are civic and share none.
4. **The number of wells orders the models.** The pair-sharing rate inverts to a well count W-hat: Qwen 39, Mistral 56, Haiku 165 to 215 at the story layer; Haiku 5.4 and Sonnet 14.6 at the theme layer. Brad's speculation of 2026-11-05 is that W-hat rises with model scale within a family; the Sonnet, Opus and Fable 100-seed cells would test it.
5. **Seeds leave a trace, and essays do not fall into wells.** The same seed in two samples shares a well 17 times more often than chance (9 of 100 pairs against 0.5), so the seed tilts the choice among the model's wells. An essay loop keeps its subject for 100 steps and converges in stance instead: by step 99, 0.85 of essays argue a position, from 0 at the start.

Background and question

Hintze (2025) ran an image loop: an image is captioned, the caption is rendered, the new image is captioned, 100 times from 100 seed prompts. The trajectories converged on a handful of attractors (a cathedral, a mirror lake, dancers) regardless of seed. We replicated that with his code and seeds and extended it across captioners and generators (reports A0, E1, E2, E6 in the repository). The attractors belonged to the generator in image space and to the captioner in caption space.

The question here is whether a loop with no image has attractors at all, and if so whose. The text loop keeps Hintze's shape: a story is compressed to a summary, the summary is expanded to a new story, and the pair repeats. The summary is the bottleneck, as the caption was. Three widths were run: 50 words (the core arm), 20 words, and a one-sentence theme of at most 25 words with no names. A rewrite loop with no bottleneck (story to story) and an essay loop (encyclopaedia topics instead of story seeds) were run as controls.

The seeds are Hintze's 100 image prompts, used as story prompts, so every cell starts from the same 100 texts and the text results sit next to the image ones.

Method

Loop. Step 0 is a story of about 500 words written from the seed prompt. Each step then asks the model for a summary of at most N words (N = 50, 20, or one sentence of at most 25 words with no character or place names) and then for a new story of about 500 words from that summary alone. The model sees only the summary, never the earlier story. 100 steps per trajectory.

Models. Claude Haiku 5.5 and Sonnet 5.5 over the API at the default sampling (the API rejects a temperature for these models), effort low; Mistral 7B and Qwen2.5-VL 7B (text only) through Ollama on an RTX 5070 laptop at temperature 0.9; LLaVA 7B in the first round. Every model tag is pinned by digest and logged in each record.

Measurement. Stories and summaries are embedded with MiniLM (all-MiniLM-L6-v2). A second embedding replaces every proper name with X, because the models reuse names (Maya, Ruth, Elara) and a shared name is not a shared plot. An atlas is k-means over all story steps of a cell with k = 40 per 2,000 steps, cluster radius 0.15, exit at 1.25 times the radius. A well is a cluster that a trajectory stays inside for at least 5 consecutive steps; a shared well is one with two or more such trajectories. The pair-sharing rate is the number of trajectory pairs that share a well over all pairs, and W-hat, its inverse, is the size of the

underlying well menu. Cells are compared by projection: the other cell's steps are labelled against a fixed reference atlas with no refit, and a trajectory is in a reference well if it stays there for 5 steps. The reference for Haiku is the pooled 200-trajectory atlas (51 shared wells).

Judges. A Haiku judge codes each story on a fixed schema (lead age and gender, lead role, setting, conflict type, ending valence, secondary character, three presence flags). A second Haiku judge scores a story against the five morals of the Haiku theme-bottleneck cell or none. The judges see no seed, step, model or well. Nothing that steers the loop is used to grade it: the loop's own summaries are not the measurement, and every atlas is built on MiniLM, which no loop uses.

Discipline. Each cell has a prediction written in the repository before it runs, never edited after. Gate reports grade the prediction as written, then say what we read into the result. Hand reads of a cell's wells are recorded before the projection or judge result is looked at. Runs are append-only and resume after a crash.

Finding 1: wells appear at 100 seeds and obey the birthday law

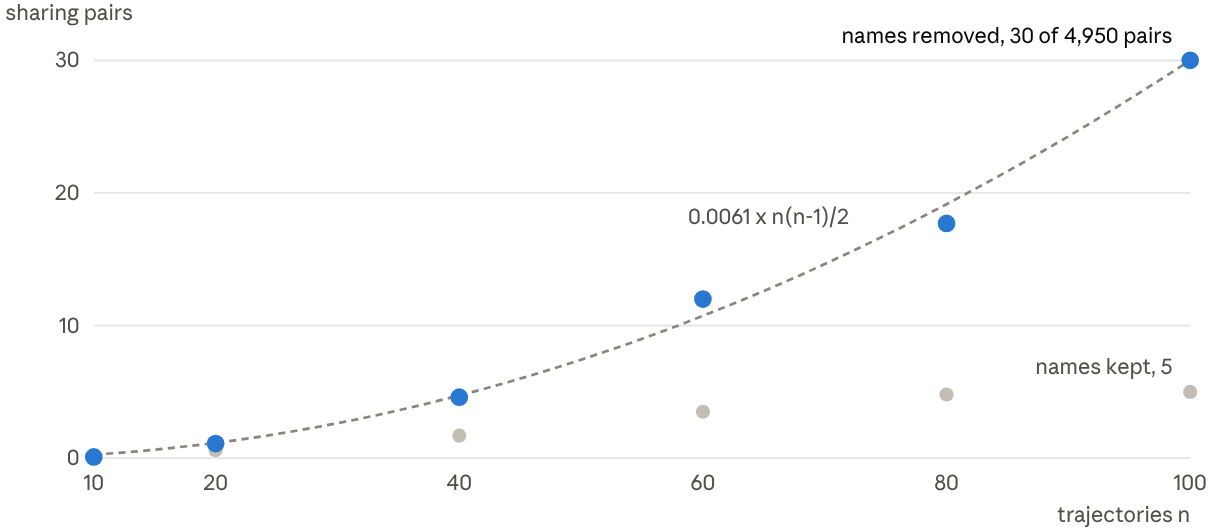
At 20 seeds nothing is shared. In six 20-trajectory cells (Sonnet, Qwen, LLaVA, each with and without the bottleneck) no persistent cluster held more than one trajectory, and the first prediction, written on the image result, failed on every clause. The loop was not converging across trajectories; it was holding each trajectory on its own path, with the seed still identifiable at step 50 in 0.45 of Haiku trajectories.

At 100 seeds the wells appear. The anonymised Haiku atlas has 13 clusters shared by two or more trajectories, 30 sharing pairs of 4,950, two wells with four trajectories each. The per-pair rate is the same at every subsample size from 10 to 100 (0.006), so the number of sharing pairs grows with the square of the number of trajectories, which is the birthday arithmetic for n draws from a menu of W-hat equally likely wells, with W-hat about 165. The 20-seed cells were not wrong, only small: at n = 20 the expected number of sharing pairs is one.

A second 100-seed Haiku sample with fresh step-0 stories replicated the law (rate 0.0046, 17 shared wells, W-hat 215) and recurred on the first sample's wells: 7 of 13 first-sample wells are visited persistently by second-sample trajectories when the second sample is projected onto the first atlas, 6 of 13 under a joint refit. Four wells were new (performers who freeze on stage, an exposé that collapses). The pooled 200-trajectory atlas has 51 shared wells and is the Haiku reference for every later comparison.

Sharing pairs grow with the square of the number of trajectories

Haiku 100-seed cell, subsamples of n trajectories; per-pair rate constant at about 0.006 sharing pairs



reports/e3/scale_width.json, pairs_vs_n: mean sharing pairs over 10 random subsamples per n (1 at n = 100), E3-S gate 2026-11-02

The points with names removed follow the dashed n-squared line; with names kept the count flattens at 5 because a shared name without a shared plot does not pass the radius.

Finding 2: the wells are plots, and the plots belong to the model

A well is as identifiable as Hintze's cathedral once it is read. Haiku's 51 pooled wells are domestic object-plots, each built around a thing left behind and the relative who left it.

Haiku well (anonymised)	sample 1 (100 seeds)	sample 2 (100 fresh step-0 stories)
A shoebox of unopened letters from an estranged relative, read after the funeral; reconciliation follows	two wells of 4 trajectories	2 trajectories stay, 2 more touch
A late father's ledger or notebook of forgiven debts and secret kindnesses; the town rallies	two wells of 3	3 stay, 1 more in the notebook well
Two estranged sisters reconcile after their father's death	one well	4 trajectories
A pianist freezes and weaves in the mother's unfinished lullaby	one well	touched by 1

Haiku well (anonymised)	sample 1 (100 seeds)	sample 2 (100 fresh step-0 stories)
A performer freezes on stage and is carried through	absent	new well

The 7B models have wells of their own kind. Mistral's 35 wells are one civic macro-plot, a stranger revives a decaying town, differing in the instrument (a mural, a concert, a bakery, a bookstore). Qwen's 33 wells are the same plot with a different agent: a gardener, a planner or a young artist turns a neglected lot, park, manor or schoolhouse into a community hub, often with a sustainability programme that goes national. Neither 7B menu has a grief well, a family-secret well or an object-plot, and both end hopeful almost without exception.

Projection makes the separation a number. Of 100 Mistral and 100 Qwen trajectories, none stays in any of the 51 Haiku wells, and their late steps sit 0.46 from the nearest Haiku well against a radius of 0.15. The two 7B models do enter each other's wells: 7 Qwen trajectories stay in Mistral wells and 10 Mistral trajectories in Qwen wells, at distances of 0.30 and 0.28. Within the Claude family the picture reverses: 2 of 20 Sonnet trajectories at 50 words sit in Haiku wells, the estranged sisters and the ledger, the same two the Haiku 20-seed cell uses.

The judge sees the same split in the stories' shape. Haiku's late stories are led by a woman 0.67 of the time, with internal conflict 0.44, home interior 0.25 and a sibling or parent as the second character; Mistral's leads are of unclear age 0.86 of the time, in social conflict 0.46, in a city 0.36, with an artist's hands on the plot.

Finding 3: the theme bottleneck removes the plot and keeps the moral

Narrowing the summary changes what survives. At 50 words the seed is identifiable for a median of 11 steps and a trajectory dwells in a well for 4 to 5 steps at a time. At 20 words the seed half-life halves to 5, but the freed trajectories do not move toward one another. Under a one-sentence theme with no names, the seed is gone in one step and the raw embedding never dwells: each trajectory visits 7 to 12 clusters in 100 steps with a median stay of one step, and two-thirds of steps are outside every cluster. A 25-word moral re-rolls the plot every step.

What it does not re-roll is the moral. The Haiku moral judge, scoring late stories against the five morals read from the Haiku theme cell, finds that the median Haiku trajectory keeps 0.79 of its late stories on one moral, and 16 of 20 trajectories keep at least 0.6. The loop has no plot memory and a strong moral memory. Each model converges to a stationary menu of about five morals.

moral (Haiku theme cell, anonymised wells)	Haiku share	Sonnet share
Kindness ripples outward through strangers, each passing it on	0.05	0.40
Grief is eased by small kindnesses from a community	0.10	0.17
Grief is softened through family rituals and shared memory	0.15	0.15
Estranged relatives reconcile; pride gives way to humility and honesty	0.52	0.11
A shared crisis dissolves old grudges between neighbours	0.07	0.00
none of the five	0.12	0.17

Sonnet's theme cell, run as the within-family test, lands 0.82 of its stories on Haiku's five morals, near the 0.89 that Haiku's own stories reach, with a hand read made before the judge putting three of Sonnet's top five morals in Haiku's five. The two menus are one set with different weights: Haiku leads with estranged siblings, Sonnet with the stranger whose kindness returns. The morals Sonnet adds are held by single trajectories (an unseen keeper whose reward is the uneventful day; accepting help as belonging; science combined with inherited local knowledge). Mistral's theme cell is civic and shares none of the five: an artist's work revives a decaying town in 5 of its 8 wells.

Finding 4: the number of wells orders the models

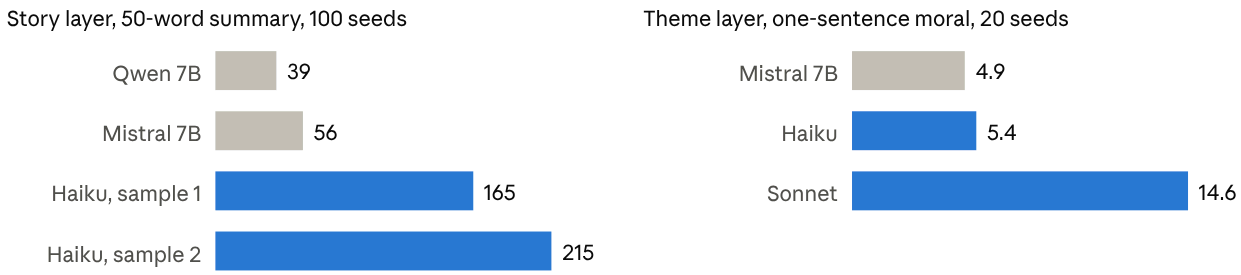
The first prediction for the 7B cells, that they would share at the same rate as Haiku, failed in a specific direction: both 7B cells share more per pair (0.018 and 0.026 against 0.005) through fewer wells. The rate is pairs per well menu, so its inverse W-hat is the size of the menu, and on W-hat the models separate cleanly at the story layer: Qwen 39, Mistral 56,

Haiku 165 to 215. Qwen's largest well holds 7 trajectories where Haiku's largest holds 4, and Qwen's wells are shared even with names kept (10 raw wells against Haiku's 4 to 5) because the same names recur.

The theme layer gives the same order within the Claude family. The Haiku theme cell has W-hat 5.4, the Sonnet theme cell 14.6, and the difference is a flatter menu rather than longer stays: Haiku parks half its trajectories on one moral, Sonnet's top moral carries 0.40 and five of its morals are held by one trajectory each.

The number of wells (W-hat) rises from the 7B models to Haiku and from Haiku to Sonnet

W-hat = trajectory pairs / pairs that share a well; the Claude family in blue



reports/e3_s7b.md, e3_s2.md, e3_ds.md: anonymised own atlases, k = 40 per 2,000 steps

Brad's speculation, recorded on 2026-11-05 before any Sonnet or Opus 100-seed cell: the models of one family share a menu, and the number of wells rises with model scale. The story-layer line from the 7B models to Haiku and the theme-layer step from Haiku to Sonnet both point that way, with two caveats. The 7B models are a different family from Haiku, so the story-layer comparison mixes family with scale, and the theme-layer comparison is at 20 seeds. The test is the 50-word loop at 100 seeds on Sonnet, Opus and Fable, which is what the credit requests ask for.

Finding 5: seeds leave a trace, and essays do not fall into wells

The second Haiku sample used the same 100 seeds with fresh step-0 stories, which made a same-seed test possible. If wells were chosen independently of the seed, 0.5 of the 100 same-seed pairs would share a well at the pooled rate. Nine did. The seed does not decide the well, but it tilts the choice among the model's wells: a seed about a lullaby tends to the unfinished-melody well, a seed about a note tends to the letters. Later cells therefore use fresh seeds per repeat or report same-seed pairs separately, and the same-seed check between the two 7B cells and between each 7B cell and Haiku found 0 of 100 pairs sharing a well, so the 7B wells are the models', not the seeds'.

The essay loop is a different instrument. Twenty encyclopaedia topics summarised to 50 words and re-expanded as essays keep their subject for the whole run: the seed is identifiable in 0.80 of trajectories at step 99, against 0.13 for stories, and no well is shared. What converges is stance. The share of essays that argue a position rises from 0 at step 0 to 0.85 at steps 50 to 99, the share written at an abstract level from 0.15 to 0.60, and the share with a hopeful tone sits at 0.55. A bias arm on essays needs a thesis bottleneck, one sentence of claim, rather than a 50-word summary.

Limits, failed predictions and next cells

What failed as written. The first text prediction (convergence as in the image loop) failed on every clause at 20 seeds. The scale prediction held on the law and missed on half-life. The replication's well-recurrence clause missed by one well under the pre-registered joint refit (6 of 13) and held under projection (7 of 13), which is why projection onto a fixed atlas is the matching method from then on. The 7B level prediction failed upward on rate and downward on W-hat. The Sonnet theme cell's dynamics and shape clauses failed on degree. Every failure is in a gate report next to the clause it failed.

Limits. One embedding (MiniLM) and one radius (0.15) define a well; the atlas is k-means, so wells near a radius boundary can split or merge between fits, which the projection method and the pooled reference contain but do not remove. The judges are Haiku, and the moral judge scores Sonnet against a Haiku menu; the hand reads made before each judge result are the check on that. The 7B cells ran at temperature 0.9 while the Claude cells ran at the API default, so the family comparison carries a sampling difference the image loops did not have. Costs were small (about $150 of API calls for the Claude cells and about 110 laptop-hours for the 7B cells), which is also the reason the Claude family above Haiku is measured only at 20 seeds.

Next cells, predictions written first. The 50-word loop at 100 seeds on Sonnet (about $145), Opus (about $290) and Fable (about $700), to measure W-hat against scale within one family with the sampling held constant; a thesis-bottleneck essay loop for the bias arm; the theme cell on Qwen. The image-loop intervention (a D&D-tuned captioner and generator, E5) runs in parallel on a pod.

Appendix: the cells

All cells use Hintze's seed prompts in order (seeds 0 to 19 or 0 to 99), 100 steps, stories of about 500 words. Rate and wells are from each cell's own anonymised atlas at k = 40 per 2,000 steps. Half-life is the median step at which the seed is no longer identified among

the cell's trajectories. Compute is API cost for the Claude cells and laptop hours (RTX 5070, Ollama) for the 7B cells.

cell	model	seeds	bottleneck	half-life	pair-sharing rate	shared wells	W-hat	compute
e3_1_sonnet	Sonnet 5.5	20	50 words	45	0	0		$27
e3_1_qwen	Qwen2.5 7B	20	50 words	14	0	0		8 h
e3_1_llava	LLaVA 7B	20	50 words	22	0	0		9 h
e3_1_mistral	Mistral 7B	20	50 words	22	0	0 (1 with names kept)		7.5 h
e3_1_haiku	Haiku 5.5	20	50 words	11	0	0		$1.30
e3_1w20_haiku	Haiku 5.5	20	20 words	5	0.011	2		$1.30
e3_1_haiku_100	Haiku 5.5	100	50 words	12	0.0061	13	165	$6.50
e3_1_haiku_100b	Haiku 5.5	100	50 words	10	0.0046	17	215	$6.50
e3_1_mistral_100	Mistral 7B	100	50 words	12	0.0178	35	56	36 h
e3_1_qwen_100	Qwen2.5 7B	100	50 words	8	0.0255	33	39	38 h
e3_d_haiku	Haiku 5.5	20	one-sentence theme	1	0.184	7	5.4	$1.30
e3_d_mistral	Mistral 7B	20	one-sentence theme	2	0.205	8	4.9	7 h
e3_d_sonnet	Sonnet 5.5	20	one-sentence theme	1	0.068	5	14.6	$26
e3_e_haiku	Haiku 5.5, essays	20	50 words	none (0.80 identified at 99)	0	0		$1.50
e3_2 and e3_2b (six cells)	Sonnet, Qwen, LLaVA, Haiku, Mistral	20	none (rewrite)	none	0	0		$27 and laptop

The Haiku theme judge ran on every cell (about $0.47 per 1,000 stories); the moral judge on the Haiku 100-seed cell and the Haiku and Sonnet theme cells. Gate reports, in order: e3, e3_followups, e3_scale_width, e3_d, e3_p, e3_s2, e3_e, e3_ds, e3_s7b, all under reports/ in the ai_to_ai_loops repository with the predictions file, the run records and the judge archives.